

VELMÈRE PRO AUDIT REPORT

Polkadot Relay Chain (native-polkadot-relay)

VELMERE SECURITY AUDIT REPORT: POLKADOT RELAY CHAIN

Audited Contract: native-polkadot-relay
Network: Polkadot Relay Chain (Chain ID: 392)
Report ID: vlm_dot_pro_366 | Tier: PRO | Application Surface: BROWSER
Created At: 2026-09-07T21:38:33.424Z

VERDICT SUMMARY: NOT SCORED (MISSING BYTECODE)
Confidence Score: 0/100 | Evidence Coverage: 0%
Bytecode is missing or malformed. In accordance with Velmère truth invariants, bytecode-derived claims are NOT emitted and security risk is marked NOT SCORED.

--- PROTOCOL OVERVIEW & NATIVE L1 NETWORK CONTEXT [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Summary of distributed architecture and consensus mechanism
Protocol / Asset: Polkadot Relay Chain
Architecture Type: Native Base Layer (Layer-1 · No Smart Contract)
Network Identifier: native_chain:dot
Native Symbol: DOT
Rule Governance: Distributed node consensus across validators / miners

--- CONSENSUS SPECIFICATION & NODE CLIENT VERIFICATION [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Analysis of block validation rules, transactions, and cryptographic curves
Verification of Polkadot Relay Chain is conducted against native Layer-1 consensus specifications. EVM bytecode decompilation is NOT_APPLICABLE for native base protocol assets.

--- BASELINE NETWORK SECURITY FINDINGS [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Identified risk vectors in base layer architecture

--- CONSENSUS GOVERNANCE & PROTOCOL UPGRADE ANALYSIS (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Decentralization evaluation, validator distribution, and improvement proposal process
Network Consensus Governance: Distributed Node Consensus [VERIFIED]
Smart Contract Admin Keys: NOT_APPLICABLE (Native Layer-1 Protocol) [VERIFIED]
Emergency Network Pause Capability: NOT_APPLICABLE (Decentralized Network) [VERIFIED]
Arbitrary User Balance Drain: NOT_APPLICABLE (Cryptographic Keys Enforced) [VERIFIED]
Protocol rules for Polkadot Relay Chain are enforced by distributed node consensus; no privileged smart contract owner key exists (NOT_APPLICABLE).

--- SPOT MARKET DEPTH, EXCHANGE RESERVES & NETWORK VELOCITY (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Global market liquidity, supply velocity, and mempool dynamics
Market Reserve Liquidity: Global Spot Market Liquidity [VERIFIED]
DEX AMM Pairs: NOT_APPLICABLE (Native Base Asset) [NEUTRAL]
Transfer Tax / Sell Fee: NOT_APPLICABLE (Standard Network Fee Model) [VERIFIED]
Liquidity for Polkadot Relay Chain is anchored in global spot venues and custodial settlement networks; automated market maker liquidity locks are NOT_APPLICABLE.

--- NETWORK ATTACK VECTORS & CONSENSUS RESILIENCE (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Modeling of chain reorganizations, Sybil attacks, and P2P partitioning

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External Oracle Dependency: NOT_APPLICABLE (Protocol Native Settlement) [VERIFIED]

Flash Loan Attack Vector: NOT_APPLICABLE (Non-EVM Execution Environment) [VERIFIED]

Double-Spend Attack Vector: Secured by network consensus [VERIFIED]

--- STATE ENGINE DETERMINISM & MULTI-CLIENT WIRE COMPATIBILITY (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Binary compatibility verification and state transition engine determinism

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Reference node client reproducible build verification against repository tags
 - * Consensus serialization and wire protocol binary compatibility analysis
 - * State transition engine deterministic execution boundary checks
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- PROTOCOL HARD/SOFT FORK EVOLUTION & INVARIANT PROOFS (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Tracking of consensus rule evolution and backward compatibility guarantees

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Protocol upgrade proposal activation and deployment history
 - * Backward compatibility verification of block validation logic
 - * Formal verification: NOT EXECUTED (Formal verification engine not commissioned)
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- HUMAN ANALYST REVIEW & PROTOCOL ARCHITECTURE AUDIT (ADVANCED) [ADVANCED] ---

VERIFICATION TYPE: INDEPENDENT MANUAL AUDITOR REVIEW

Attestation by independent distributed systems auditor

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Analyst sign-off requires verified reviewer evidence
 - * Cryptographically signed analyst attestation digest
 - * Manual review boundary explicit: no guarantee of absolute perfection
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
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